Einstein spectral pinching and a monopole consequence

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Einstein spectral pinching: automatic amplitude and a monopole consequence

23 September 2026. Conventions: nonnegative scalar Laplacian, Hilbert–Schmidt norm on the Weyl endomorphism, and exterior norm on forms. This note supplies two applications of the harmonic-Weyl selection theorem. The maximum-principle calculation is elementary; no independent priority is claimed for that calculation. The quantitative topological consequence uses the accompanying selection estimate, with its complete hypotheses.

1. An upper amplitude supplied by the equation

Let (M, g) be a closed oriented Einstein four-manifold with $\mathrm{Ric} = -g$. Write the ordered eigenvalues of W^+ as $\lambda_1 < \lambda_2 \leqslant \lambda_3$, and put

$$t = -\frac{3}{4}\lambda_1 > 0, \qquad b = \lambda_3 - \lambda_2, \qquad \theta = \frac{b}{4t}. \tag{1}$$

The lowest eigenvalue is assumed simple everywhere. The function t is smooth, while b need only be continuous at an upper-eigenvalue collision. Its square is smooth. Trace-freeness gives

$$(\lambda_1, \lambda_2, \lambda_3) = \left(-\frac{4t}{3}, \frac{2t}{3} - \frac{b}{2}, \frac{2t}{3} + \frac{b}{2}\right), \quad |W^+|^2 = \frac{8}{3}t^2 + \frac{1}{2}b^2. \tag{2}$$

In particular, simplicity of the lowest eigenvalue is equivalent to $\theta < 1$.

Lemma 1. If $\theta \leqslant \theta_0 < 1/\sqrt{3}$, then

$$\boxed{t \leqslant M_{\theta_0} := \frac{1}{2(1 - 3\theta_0^2)}.} \tag{3}$$

There is no separate upper Weyl-curvature hypothesis.

To prove this, use the Einstein Weyl equation

$$\nabla^*\nabla W^+ = 2W^+ + 6(W^+)^2 - 2|W^+|^2 I. \tag{4}$$

For a smooth simple eigenvalue of a self-adjoint bundle section, the second derivative formula, with the nonnegative Laplacian convention, is

$$\Delta\lambda_1 = \langle(\nabla^*\nabla W^+)e_1, e_1\rangle - 2\sum_{j=2}^{3} \frac{|\langle \nabla W^+ e_1, e_j\rangle|^2}{\lambda_1 - \lambda_j}. \tag{5}$$

An orthonormal eigenbasis at the point suffices; the two upper eigenvalues need not be locally simple. Since each denominator is negative, (2)–(5) imply

$$\Delta t \leqslant 2t - 4t^2 + \frac{3}{4}b^2 \leqslant 2t - 4(1 - 3\theta_0^2)t^2. \tag{6}$$

At a maximum of t, one has $\Delta t \geqslant 0$. Equation (3) follows. The conclusion is only an upper bound: this argument does not provide a uniform positive lower bound for t over a family of manifolds.

2. A derivative-free structure estimate

Assume in addition $t \geqslant m > 0$, with a prescribed lower bound. Let ω be the normalized lowest eigenform, $|\omega|^2 = 2$, on the orientation cover of its real eigenline. The cover has degree at most two. Normalized integrals are unchanged under this cover. The harmonic-Weyl selection theorem, applied using (3), gives

$$\boxed{\fint \left(|dt|^2 + (t - \tfrac{1}{2})^2 + |\nabla\omega|^2 \right) \leqslant C_{m,\theta_0} \fint b^2.} \tag{7}$$

Here the Einstein equation supplies harmonicity of W^+ and $s = -4$. No assumed Hessian budget, Ricci derivative estimate, Sobolev constant, injectivity radius, scalar spectral gap, diameter, or W^- bound is present. The lower amplitude and relative spectral pinching are genuine assumptions. This is an integral structure estimate, not a theorem constructing a nearby Kähler metric.

3. Topology forces a definite upper-eigenvalue splitting

Suppose that the given metric admits an actual polarized monopole class with $p = (a_g^+)^2 \geqslant A > 0$. This can be the metricwise class on the blow-up/handle families of the main integral theorem; no globally fixed class working for all polarizations is being presumed. Put

$$T = 2\chi(M) + 3\tau(M), \qquad h = 1 - T/A, \quad 0 < h < 1. \tag{8}$$

Corollary 2. Under the hypotheses of Section 2 and (8),

$$\boxed{\fint_M (\lambda_3 - \lambda_2)^2 \geqslant c_{m,\theta_0} h^2.} \tag{9}$$

It applies to every normalized negative Einstein metric meeting these spectral assumptions, not merely to a selected construction.

Indeed, the scalar monopole estimate and $s = -4$ give

$$16V \geqslant 32\pi^2 p \geqslant 32\pi^2 A. \tag{10}$$

For an Einstein metric, Chern–Weil gives

$$2\pi^2 T/V = \frac{1}{3} + \fint |W^+|^2. \tag{11}$$

Since $T = A(1 - h) > 0$, (10)–(11) imply

$$\fint |W^+|^2 \leqslant \frac{2}{3} - h. \tag{12}$$

Combining (2) and (12), and discarding only nonpositive terms, yields

$$h \leqslant \frac{8}{3}\left(\frac{1}{4} - \fint t^2\right) - \frac{1}{2}\fint b^2 \leqslant \frac{8}{3}(M_{\theta_0} + \tfrac{1}{2})\left(\fint (t - \tfrac{1}{2})^2\right)^{1/2}. \tag{13}$$

Use (7) and square this inequality. This proves (9), with, for example, $c_{m,\theta_0} = [\frac{8}{3}(M_{\theta_0} + \frac{1}{2})]^{-2}C_{m,\theta_0}^{-1}$, where the latter constant is the one in (7).

On $M = X\#k\overline{\mathbb{CP}}^2\#\ell(S^1\times S^3)$, $A = c_1^2(X) > 0$, the parameter is $h = (k+4\ell)/A$. Thus the inequality detects a required spectral departure from the negative Kähler–Einstein curvature pattern on this actual family. It does not exclude metrics with a lowest-eigenvalue collision, very small lower amplitude, or failure of the displayed pinching. Consequently this corollary alone is not an unconditional Einstein nonexistence result. The separate two-chirality theorem supplies the new unconditional topological application in this research package.

4. Sources and what is being added

The Einstein Weyl equation is classical; see Derdziński, Self-dual Kähler manifolds and Einstein manifolds of dimension four, and the formulas recalled in Catino–Dameno, Conformal Kähler rigidity of Einstein four-manifolds. The latter paper's largest-simple-eigenvalue classification must not be replaced by a lowest-simple-eigenvalue classification: the negative Kähler–Einstein model has a simple lowest eigenvalue and a double largest one. The accompanying harmonic-Weyl proof derives the signed balance for this different branch.

The scalar monopole estimate is the classical input of LeBrun, Ricci curvature, minimal volumes, and Seiberg–Witten theory. Equations (10)–(12) are classical consequences. The present quantitative application is their composition with (7); it is not a new proof of the classical inequalities or a complete spectral classification of Einstein four-manifolds.